

Article

Modeling the Discharge Rate of a Screw Conveyor Considering Hopper–Conveyor Coupling Parameters

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Abstract: Developing a flow rate model for the screw feeder and optimizing discharge performance are crucial for achieving automated intelligent precision feeding. This study constructs a mass flow rate model for screw conveyors, considering the coupled structural parameters of the hopper and screw conveyor. The model is developed using single-factor tests and central composite design (CCD) response surface tests and is validated through actual discharge tests. Results indicate that the discharge rate in the hopper–screw conveyor system is primarily influenced by the screw conveyor itself. Among the structural parameters, the hopper inclination angle and the hopper discharge opening length significantly affect the filling coefficient. Validation tests show an average error of 6.8% between the predicted and simulated mass flow rates and 5.0% with the actual mass flow rate, demonstrating the model’s high precision and accuracy.

Keywords: hopper; screw conveyor; discharge rate model; DEM; central composite design



Citation: Chen, P.; Huang, T.; Wu, B.; Qian, H.; Xie, F.; Liu, B.; Liu, D.; Li, X. Modeling the Discharge Rate of a Screw Conveyor Considering Hopper–Conveyor Coupling Parameters. *Agriculture* **2024**, *14*, 1203. <https://doi.org/10.3390/agriculture14071203>

Received: 16 June 2024

Revised: 5 July 2024

Accepted: 19 July 2024

Published: 22 July 2024



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1. Introduction

A screw feeder consists of a hopper and screw conveyor, it is a kind of machine that uses its motor to drive a screw blade that rotates and pushes the material to realize the purpose of conveying; its structure is simple, easy to install and maintain, and it is widely used in various trades and professions (agriculture, mining, chemical industry, food industry, etc.). Developing a flow model of the discharge parts of the screw feeder and the optimization of the discharge performance is of great significance for the realization of automated intelligent and precise feeding.

In the study of flow rate models for hoppers, Beverloo [1] first proposed a hopper discharge flow rate model, and subsequent flow rate models were basically based on this model with modifications. Brown and Richards [2,3] modified the model according to the minimum energy theory and the free-fall arch theory by considering the influence of the semi-apex angle of the hopper, and Myers and Sellers [4] established a flow rate model for a wedge-shaped hopper by considering for the first time the influence of the change in hopper geometry on the flow rate. Subsequent studies focused on different discharge scenarios, and the modification of the discharge flow model took into account various factors during discharge, such as van der Waals forces between particles [5], internal friction coefficients between particles [6], dynamic particle-particle contact interactions [7], particle size distribution [8], etc., and also considered the effect of the hopper geometry and the insert in the hopper [9,10]. In general, these flow rate models corrected one or more parameters in the existing flow rate model by considering some new discharging

conditions or inserted new parameters into the existing flow rate model to expand the scope of application of the model.

For the modeling of the flow rate of the screw conveyor, the empirical formula shown in Equation (4) is usually used to predict the flow rate. This model takes the diameter of the spiral blade, pitch, rotation speed, material filling coefficient, inclination angle, material bulk density, and other factors into consideration. The modification of this model has been studied mainly considering the variation in two parameters, pitch and volumetric efficiency. Aiming at the issue of predicting the flow rate of a variable-pitch screw conveyor, reference [11] proposed a flow rate calculation method on the basis of the above flow rate model. Volumetric efficiency, which is the material filling coefficient, was defined as the ratio of the volume of the material to the volume of the geometry space within one pitch range of the screw. Reference [12] presented the relationship between screw speed and volumetric efficiency by studying the discharge of rice grain in a screw conveyor. Considering the influence of particle characteristics [13,14], particle motion [15], and variable pitch [16], different methods to predict volumetric efficiency have been proposed, which were used to correct the flow rate model of the screw conveyor. The model proposed in reference [15] can even be used to predict the volume flow rate of the screw conveyor.

In the theoretical study of the screw conveyor, experimental studies were conducted on the rotational speed [17], elevation angle [18], screw clearance [19], and other related parameters of the screw conveyor, and the results showed that the volume of the screw conveyor decreased linearly with the increase in elevation angle [20], and the smaller the ratio of the pitch to the outer diameter, the higher the volumetric efficiency, and the torque increased with the increase in feed rate after exceeding a certain value [21]. In the DEM (Discrete Element Method) simulation study of screw conveyors, the related factors such as material filling level [22,23] and material filling rate [24,25] were investigated, and the related results showed that the material filling level and rotational speed directly affected the relative velocity of particles with respect to the screw blades, and both parameters were the key factors influencing the discharge efficiency of the screw feeder [26].

In summary, there is a mature basis for related research on hoppers and screw conveyors and their flow modeling, but screw conveyors are always used in combination with hoppers. For a screw feeder consisting of a wedge hopper and a screw conveyor, the flow rate of the wedge hopper and the screw conveyor could both be predicted by readily available flow rate models. But these two models are independent, and neither takes the effect of the coupling parameters of the wedge hopper and the screw conveyor into account. Therefore, modeling the flow rate by considering the coupled structural parameters of the wedge hopper and the screw conveyor is important for accurately predicting the discharge flow rate of the screw feeder. In this paper, EDEM software (EDEM 2021.2) is used to simulate the discharging progress of the screw conveyor, and the effect coupling parameters (i.e., hopper discharge opening length, discharge opening width, hopper offset distance, hopper inclination angle, and hopper semi-apex angle) on filling coefficient of screw conveyor is explored. Then, a CCD (central composite design) response surface test is carried out to construct the regression model of the filling coefficient, and the existing flow rate model of the screw feeder is corrected according to the regression function.

2. Materials and Methods

2.1. Theoretical Background

2.1.1. Hopper Flow Rate Modeling

Initially, Beverloo [1] conducted experiments in cylindrical flat-bottom hoppers with particles of different sizes and shapes and proposed the most widely used flow rate prediction, Equation (1), which is applicable to the prediction of discharge rates for all cylindrical flat-bottom hoppers.

$$W_{Beverloo} = C\rho_b g^{1/2} (D_0 - kd)^{5/2} \quad (1)$$

where $W_{Beverloo}$ is the flow rate of the material, g is the acceleration of gravity, D_0 is the diameter of the discharge opening of the hopper, d is the mean diameter of the particles, k is an empirical constant related to the size and shape of the particles, C is Beverloo's constant.

Myers and Sellers [2] first considered the effect of changes in hopper geometry on flow rate and expressed the flow rate prediction model for a wedge-shaped hopper with a discharge opening width of W_0 and discharge opening length of L as Equation (2), which is applicable to flow rate prediction for a wedge-shaped hopper with a semi-apex angle of 90 degrees.

$$W_{Myers} = 1.03\rho_b g^{1/2} (L - kd)(W_0 - kd)^{3/2} \quad (2)$$

where L is the discharge opening length, and W_0 is the discharge opening width.

Since the wedge-shaped hopper geometric parameters affect the hopper flow rate, in Brown and Richards [3,4], according to the minimum energy theory and free-fall arch theory and considering the influence of the hopper semi-apex angle, the flow rate equation of the flat-bottom hopper was revised as in Equation (3); that model was applicable to the symmetric wedge-shaped hopper with equal left and right semi-apex angles for predicting the flow rate.

$$W_{Brown} = W \cdot \left(\frac{1 - \cos^{3/2} \alpha}{\sin^{5/2} \alpha} \right) \quad (3)$$

where W is the flat-bottom hopper discharge rate, and α is the flat-bottom hopper half-top angle.

2.1.2. Screw Conveyor Flow Rate Modeling

Screw feeders consist of a hopper combined with a screw conveyor. Influenced by material characteristic and geometric parameters, particles exhibit complex flow behavior during discharge, making it very difficult to predict the discharge rate of the screw feeder. For this reason, scientists have carried out a great deal of research into the construction of flow rate models for hoppers and screw conveyors, respectively. For the screw conveyor, the flow rate model is defined as Equation (4) [11]:

$$Q = 47D^2\rho_b\varphi SnC_0 \quad (4)$$

where Q is the screw feeder conveying capacity, D is the radius of the screw blade, S is the pitch, C_0 is the correction factor for inclined conveying, n is the rotational speed, ρ_b is the density of the material accumulation, and φ is the material filling rate.

2.1.3. Flow Rate Comparison

In order to investigate the relationship between the flow rate of the wedge-shaped hoppers, the flow rate of the screw conveyor, and the flow rate after coupling, assumptions were made based on the existing flow rate model; the wedge-shaped hopper was assumed to be designed according to the minimum design specifications and the screw conveyor according to the maximum design specifications.

For the wedge-shaped hopper, to prevent clogging of the discharge, the discharge opening width of the hopper was taken as 30 mm; when it is less than this value, the particles are clogged in the process of discharging. The semi-inclined angle of the silo was preset as 60 degrees, and the length of the discharge opening was taken as 2 times the width of the discharge opening, i.e., $W_0 = 30$ mm, $L = 60$ mm.

For the screw conveyor, the spiral speed range is mainly in the range from 50 rpm to 140 rpm, and the commonly used spiral blade diameter range is $40 \text{ mm} \leq D \leq 60 \text{ mm}$ [27]. In addition, for the horizontally arranged blades, the pitch of the spiral blade diameter is generally 0.7~1.0 [28]. Thus, when designing the screw conveyor according to maximum specifications, the spiral blade diameter was taken as 60 mm, which was 2 times the wedge-shaped hopper outlet width. The pitch was taken as 1 time the spiral blade diameter. The screw speed was taken as the maximum value of 140 rpm. The filling factor of the

pellet feed was taken as 0.6 because the maximum φ value was 0.6 in the reference table of material filling factor [27]. For the screw inclination coefficient, it was taken as 1 in this study because the screw was placed horizontally.

The parameter values of the assumed parameters were substituted into the flow rate model for comparison.

$$\frac{Q}{W_{Myers}} = \frac{47D^2\rho_b\varphi SnC_0}{1.03\rho_b g^{1/2}(L - kd)(W_0 - kd)^{3/2}(1 - \cos^{3/2}\alpha)/(\sin^{5/2}\alpha)} \quad (5)$$

All the parameter values were substituted into Equation (5), and the calculations were performed to obtain the results:

$$\frac{Q}{W_{Myers}} = 0.597 \quad (6)$$

The results showed that the minimum discharge rate of the hopper was much greater than the maximum discharge rate of the screw conveyor. That is, the discharging rate of the screw conveyor coupled to the hopper was determined by the discharge rate of the screw conveyor. Therefore, in subsequent studies, the default discharge rate of the wedge hopper was much greater than the discharge rate of the screw conveyor. The granular feed fed into the screw from the hopper discharge opening was always in a “saturated” state. The coupled discharge rate could be determined from the screw conveyor discharge rate.

From Equation (4), it can be seen that the screw conveyor discharge rate model has no direct relationship with the hopper geometry parameters. However, the material filling coefficient is an uncertain quantity, which may be affected by the hopper geometric parameters and ultimately affect the discharge rate of the screw conveyor. Therefore, this study investigated the functional relationship between the relevant dimensions at the hopper–screw conveyor coupling and the material filling coefficient and then constructed a flow rate model to predict the discharge rate of the screw feeder under different coupling structures.

2.2. Simulation Condition

In this study, the “Twins” brand pellet feed was selected as the test object, and its characteristic parameters were consistent with our previous study [29]. The contact model used in the simulation software was the Hertz–Mindlinno-slip contact model [30], and the pellet feed was represented by a spherical cylindrical. The length of the pellet was set to 7.5 mm, and the radius of the pellet was 2.1 mm, as shown in Figure 1.

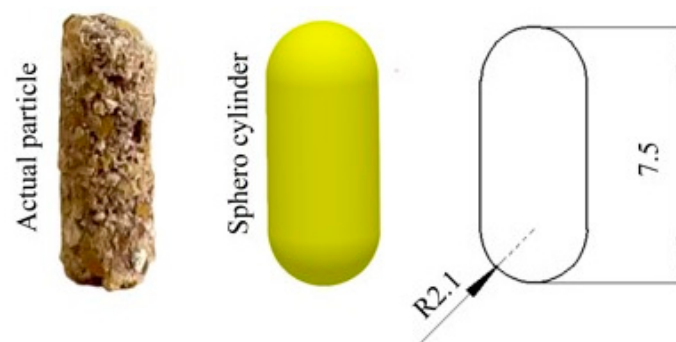


Figure 1. Granular materials.

The structure and parameters of the screw feeder in this study are shown in Figure 2; the screw feeder mainly consists of a combination of a wedge-shaped hopper and a screw conveyor so that the discharge rate of both the wedge-shaped hopper and the screw conveyor together determines the actual discharge rate of the final screw feeder.

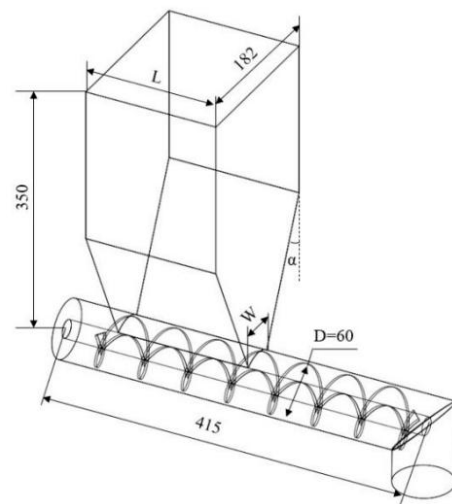


Figure 2. Screw feeder.

The geometry was imported into the simulation environment, and the mesh size was set to 5 mm. The distance between the computational domain boundary and the geometry was set to 2 mm in the +X direction, 8 mm in the +Y direction, and 2 mm in the +Z direction. The gravitational acceleration was set to 9.81 m/s^2 along the negative direction of the Y axis. The parameters relating to the particles and the wedge-shaped hoppers used in the simulation are given in Table 1.

Table 1. Material characterization parameters.

Parameters	Value	Source
Poisson ratio	0.26	Measured
Particle density (kg m^{-3})	1037	Measured
Particle dimensions $L \times R$ (mm)	7.5×2.1	Measured
Particle model	Sphero cylinder	Selected
Particle Young's modulus (Pa)	9.9×10^7	Measured
Steel Poisson rate	0.3	[31]
Steel density (kg m^{-3})	7865	
Steel Young's modulus (Pa)	7.9×10^{10}	
Particle–particle sliding friction coefficient	0.82	Measured
Particle–particle restitution coefficient	0.65	Measured
Particle–steel restitution coefficient	0.62	Measured
Particle–particle rolling friction coefficient	0.08	Measured
Particle–steel sliding friction coefficient	0.62	Measured
Particle–steel rolling friction coefficient	0.02	Measured

The particles in the wedge-shaped hopper were pre-filled, and the fill height affected the particle discharge rate if it was less than 2.5 times the discharge opening width [32]. Therefore, in order to ensure for the filling height of the particles to be consistent and independent of the filling height in any hopper, a particle factory was established 30 mm below the highest point of the hopper. Pellets were generated dynamically at a rate of 80,000 pellets per second with a generation time of 2 s. The final number of pellets varied with the shape of the hopper. The generated pellets were settled in the hopper for 2 s to put them in a stable position. Subsequently, the discharge opening was opened, and the particles were discharged under the effect of gravity. The whole simulation time lasted 60 s, and the time step of simulation was set to 5×10^{-6} .

In our previous studies [29], the above simulation model was verified by the repose angle test and discharging test. The verification results showed that the simulation model could reflect the flow of pellet feed well. In addition, in a previous study, the verification of the mesh independence and time-step independence was performed to remove the impact of the mesh size and time step on simulation outcomes, respectively. The validation results can be seen in the Supplementary Files.

2.3. Experimental Scheme

The material filling coefficient is the ratio of the cross section of material accumulation in the spiral to the cross-section area of the spiral, and its calculation in Equation (7) is deduced from Equation (4).

$$\varphi = \frac{Q}{47D^2Sn\lambda\varepsilon} \quad (7)$$

In order to investigate the effect of different factors on the filling coefficient, this experiment was carried out on the discharge opening length, discharge opening width, offset distance, inclination angle, and semi-apex angle of the hopper using five factors for each, respectively, in a single-factor test. The purpose of the single-factor tests was to screen for factors that had a significant effect on the screw conveyor filling coefficient. These tests were conducted without considering the interaction between the factors in order to reduce the number of tests and computation time.

The factor level selected for these single-factor tests is shown in Table 2.

Table 2. Single-factor level table.

Level	Inclination Angle (degree)	Offset Distance (mm)	Discharge Opening Width (mm)	Semi-Apex Angle (degree)	Discharge Opening Length (mm)
1	0	0	30	25	50
2	15	2	35	40	100
3	30	4	40	55	150
4	45	6	45	70	200
5	60	8	50	85	250

In this study, five values of each factor were taken for the single-factor test, and the initial levels of each parameter were as follows: an inclination angle of 0°, an offset distance of 0 mm, a discharge opening width of 40 mm, a semi-apex angle of 25°, and a discharge opening length of 150 mm. When a single-factor test was taken on one factor, the levels of the other factors were the initial levels. The other hopper dimensions and auger settings are shown in Figure 2. Figure 3 shows the schematic diagram of the hoppers for the selected factors. The screw speed was 40 rpm. According to the total simulation discharge time of 60 s, in the process of data processing, the data after 10 s of the stable discharge moment were selected, 4 groups were taken, and the discharge rate was the average discharge rate of these groups in 3 s in a row. The material filling coefficient could be obtained by using Equation (7).

2.4. CCD (Center Composite Design) Experiment Methods

According to the results of the single-factor test, the discharge opening length and inclination angle had a significant effect on the material filling coefficient, so a two-factor response surface test was carried out. Therefore, we utilized the CCD rotation combination center design in Design-expert software to conduct a two-factor response surface experiment and study the relationship between the above two parameters and the material filling coefficient of the screw conveyor, for a wider applicability of the screw feeder flow prediction model. The response surface test factors and levels are shown in Table 3.

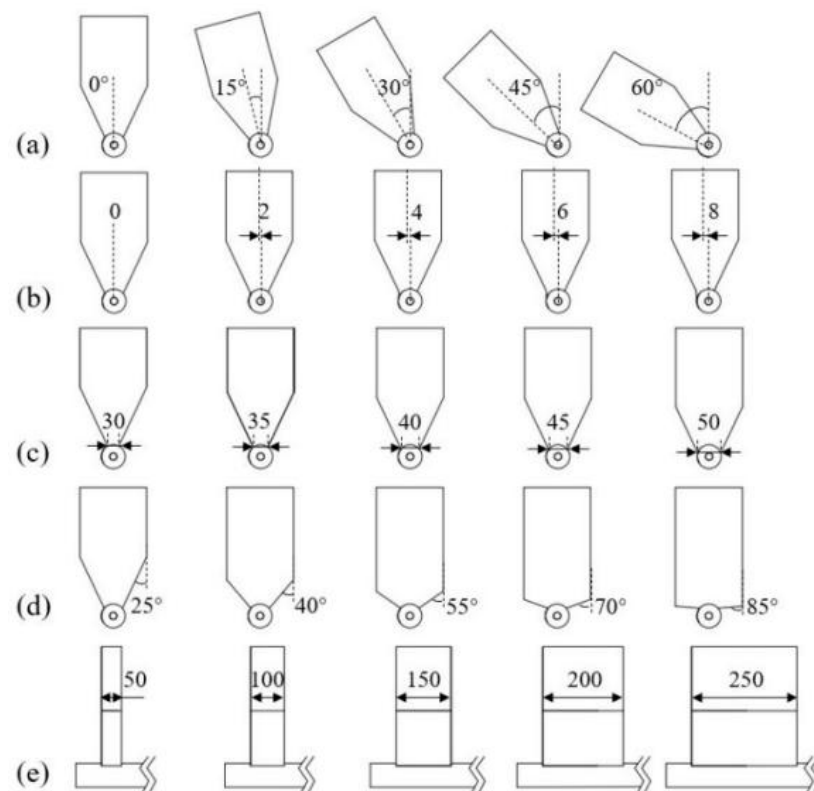


Figure 3. Single-factor levels: (a) the levels of the inclination angle; (b) the levels of the offset distance; (c) the levels of the discharge opening width; (d) the levels of the semi-apex angle; (e) the levels of the discharge opening length.

Table 3. Response surface test factors and levels.

Level	Discharge Opening Length (mm)	Inclination Angle (Degree)
−1.414	50	0
−1	79.29	8.79
0	150	30
1	220.71	51.21
1.414	250	60

The experimental design software Design-expert was utilized to design the experimental scheme, and simulation tests were carried out. According to the total simulation discharging time of 60 s, the data of the stable discharging moments from 17 s to 37 s were selected to find the average value of the discharging rate in that time period, and the filling coefficients of each group were calculated by substituting into Equation (7).

3. Results

3.1. Flow Pattern and Velocity Distribution

At the initial moment of discharging, the particles at different heights in the hopper were colored in different colors. Then, the distribution of particles in the hopper could be observed at different times during discharging, which could reflect the influence of various factors on the flow of particles. As shown in Figure 4, the discharge state of the hopper under the influence of different single factors was recorded at 2 s, 20 s, and 40 s. These three moments were the material filling completion time point and two discharging time points, respectively. The different colors represent particles at different heights at the initial moment (2 s), and the distribution of different colors reflects the flow of particles in the hopper. Figure 5 shows the velocity distribution of the particles at the outlet of the hopper under the influence of different factors at the moment of discharge at 30 s.

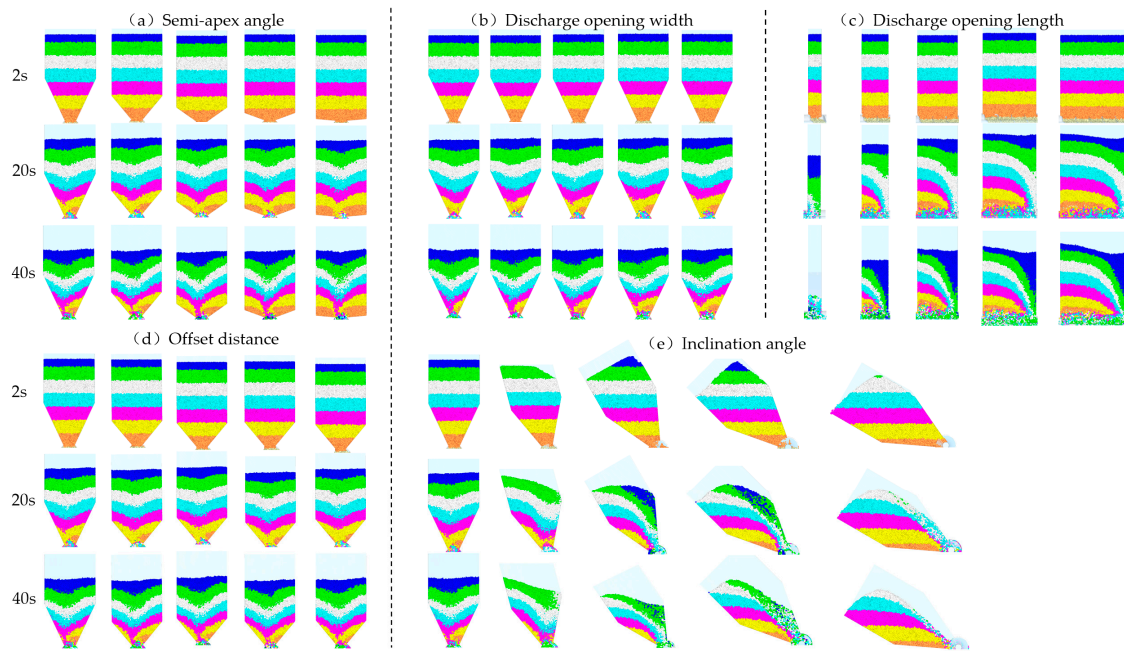


Figure 4. Flow pattern for different factors. (a) Semi-apex angles at different levels. (b) Discharge opening widths at different levels. (c) Discharge opening lengths at different levels. (d) Offset distances at different levels. (e) Inclination angles at different levels.

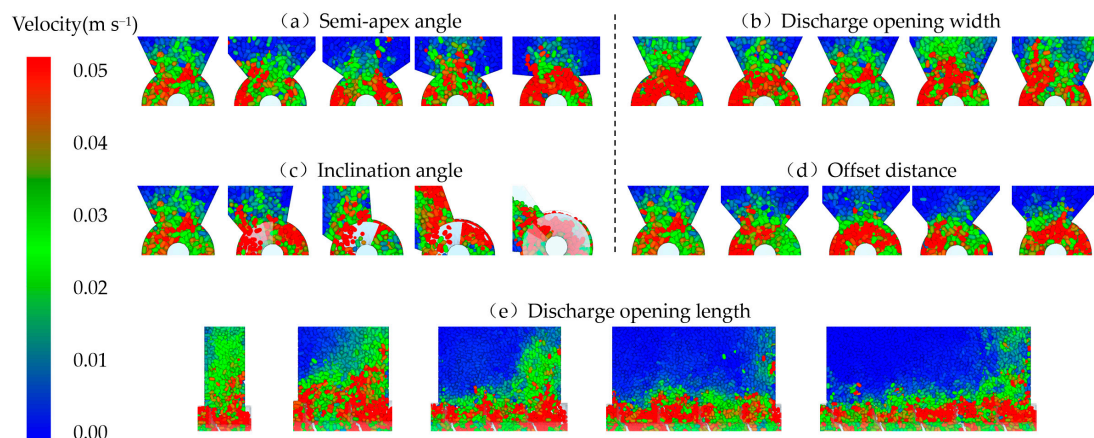


Figure 5. Velocity distribution at the outlet. (a) Semi-apex angles at different levels. (b) Discharge opening widths at different levels. (c) Inclined at different levels. (d) Offset distances at different levels. (e) Discharge opening lengths at different levels.

From Figure 4, it can be seen that the flow pattern of the wedge-shaped hopper was a mass flow in the case of changing the semi-apex angle of the hopper, discharge opening length, and the offset distance. According to Equation (6), it can be seen that the discharge process of the hopper was slow because the speed of the hopper discharge was affected by the speed of the screw conveyor during the discharge process of the screw feeder. In addition, the particles at the left bottom of the hopper were always discharged preferentially, and the “V” shape of the particle flow pattern at the bottom of the hopper tended to be shifted to the left in the latter part of the discharge, due to the clockwise rotation of the screw feeder to pick up the material from the hopper. When changing the offset angle, the particles at the top of the hopper were discharged first as the inclination of the hopper gradually increased, and at the same time, as the particles at the top of the hopper flowed out, the right side of the screw conveyor was not filled with material, resulting in a decrease in the filling coefficient. In the case of changing the discharge opening length, it can be

seen that as the discharge opening increased, the particles at the back of the hopper were discharged into the screw feeder first, while the particles at the front of the hopper remained basically static.

As can be seen in Figure 5, the velocity of the particles in the hopper increased after entering the screw conveyor, while the particles above the outlet began to accelerate, indicating that a free-fall arch still existed during the screw conveyor–bin coupling discharge process. At the same time, the velocity of the particles at the left bottom of the hopper tended to be greater than the velocity of the particles at the right bottom, indicating that the particles at the hopper outlet that were subjected to the same force as the direction of rotation of the conveyor were discharged into the screw conveyor first. When changing the discharging opening length, the velocity of the particles at the bottom of the front end of the hopper decreased and came to a standstill as the discharging opening length increased. This indicated that as the length of the hopper discharge opening increased, the auger feeder, which started filling at the rear of the hopper, continued to fill as it moved forward, and when fully filled, would cause the particles at the front of the hopper to not continue to enter the screw conveyor and come to a stationary state.

3.2. Normal Contact Force Distribution

From a microscopic point of view, the force chain generated by the mutual contact between particles forms a large and complex force network, which tends to stabilize the structure of the force chain and form a free-fall arch, thus hindering the discharge of particles above the arch. Figure 6 shows the magnitude of the normal contact force at 30 s for particles in the discharge process of the hopper under the influence of different single factors. It can be seen that when changing the hopper semi-apex angle, offset distance, and discharge opening width, the force network formed by the particles in the hopper took the form of an arch, and the normal force contact force on the right bottom part of the particles was larger than that on the left, resulting in an imbalance of force on the particles on the left and right bottoms of the hopper, which indicated that the screw conveyor tended to destroy the free-fall arch formed in the process of taking up material for discharge and at the same time made the particles on the left side enter the screw conveyor first.

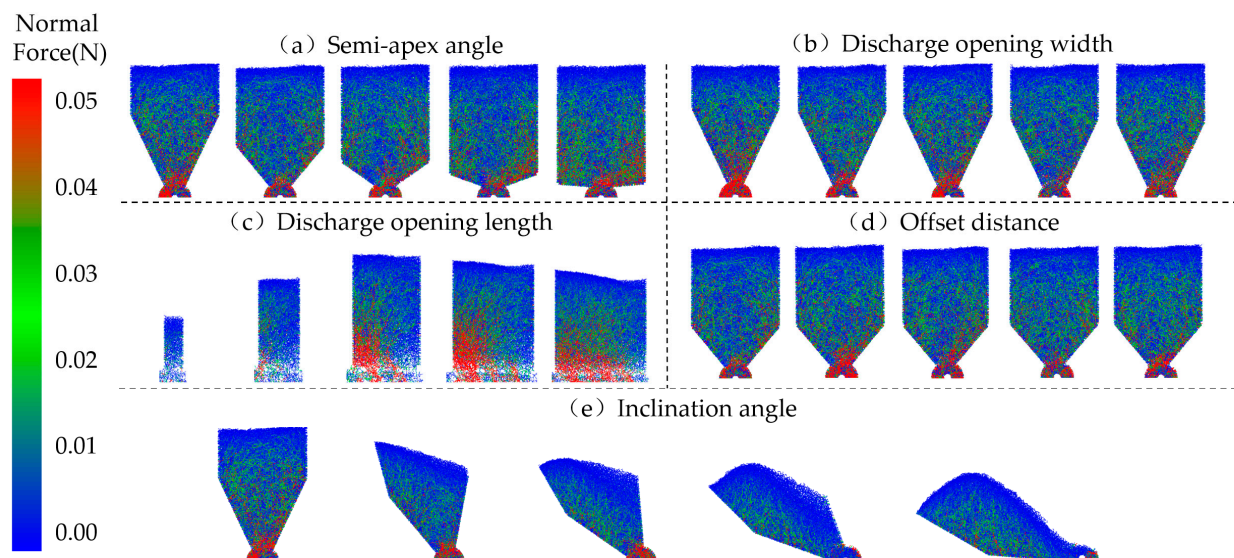


Figure 6. Normal contact force distribution at the outlet. (a) Semi-apex angles at different levels. (b) Discharge opening widths at different levels. (c) Discharge opening lengths at different levels. (d) Offset distances at different levels. (e) Inclination angles at different levels.

When changing the inclination angle of the hopper, it can be seen from the figure that with the increase in the inclination angle, the particles entering the screw conveyor were preferentially the particles at the top of the hopper, because when the inclination angle

increased, the particles in the hopper bulked up, and when the particle bulk angle was smaller than the repose angle of the material, the particles at the top slipped downward and fell into the screw conveyor, which filled only half of that part of the conveyor, leading to a reduction in the filling coefficient of the screw. This resulted in a lower fill factor in the screw conveyor. When changing the discharge opening length, as the discharge opening length increased, the screw feeder, which started to fill from the rear of the hopper, continued to fill as it moved forward, resulting in particles at the front end of the hopper not being able to enter the screw feeder. This indicated that the material at the bottom of the front end was subjected to a greater normal contact force from the particles in the already-filled screw feeder than the pressure from the upper particles, resulting in the particles at the bottom of the front end being relatively stationary.

3.3. Material Filling Coefficient

The experimental results of the material filling coefficient at different factor levels are shown in Figure 7. Moreover, the results were subjected to an analysis of variance as shown in Table 4. It can be seen that the discharge opening length and the inclined angle of the hopper had a significant effect on the results of the material filling coefficient, while the semi-apex angle of the hopper, discharge opening width, and the offset distance of the hopper did not have a significant effect on the material filling coefficient. Therefore, the relationship between the two significant influences and the flow rate model needed to be considered in the subsequent flow rate modeling.

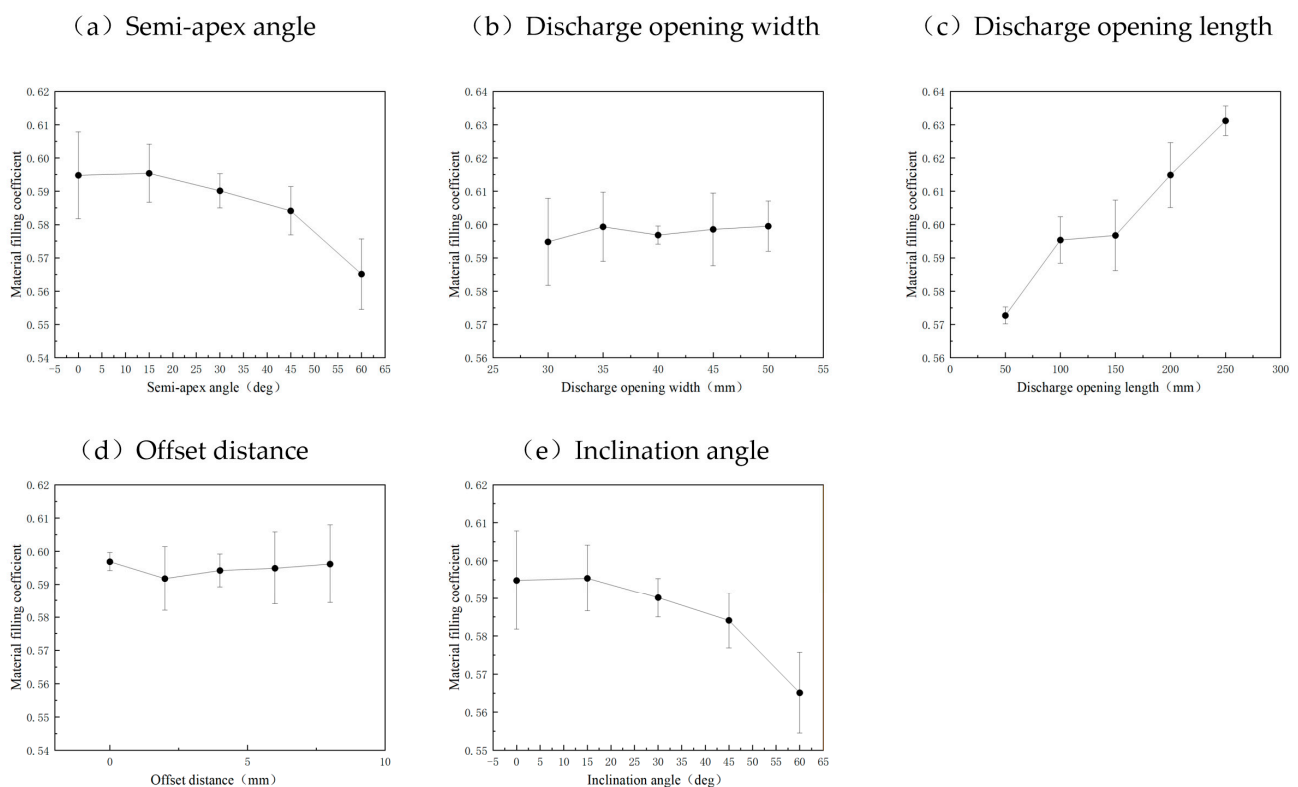


Figure 7. Results of single-factor tests. (a) Semi-apex angles at different levels. (b) Discharge opening widths at different levels. (c) Discharge opening lengths at different levels. (d) Offset distances at different levels. (e) Inclination angles at different levels.

Table 4. Analysis of variance for single factor tests.

Factors	Source of Variance	Sum of Squares	Mean of Squares	F-Value	p-Value
Inclination angle	Inter-class variance	0.00249	0.000622	7.0953	0.0021
	Between-class variance	0.00131	0.0000876		
Offset distance	Inter-class variance	0.0000617	0.0000154	0.2024	0.9331
	Between-class variance	0.00114	0.0000763		
Discharge opening width	Inter-class variance	0.000063	0.0000157	0.1703	0.9502
	Between-class variance	0.00139	0.0000924		
Semi-apex angle	Inter-class variance	0.000155	0.0000387	0.6562	0.6316
	Between-class variance	0.000884	0.0000589		
Discharge opening length	Inter-class variance	0.00779	0.00195	34.7076	0.0000
	Between-class variance	0.000842	0.0000561		

(*p*-value is the level criterion for testing whether it is significant or not; $0.01 \leq p \leq 0.05$ indicates that the factor is significant, and $p < 0.01$ indicates that the factor is highly significant).

3.4. Experimental Results and Flow Rate Modeling

The results of the response surface tests are shown in Table 5. The results were subjected to an analysis of variance as shown in Table 6, and the fitting results showed that the *p*-value of the model was <0.0001 , the *p*-value of the lack-of-fit term was $0.4004 > 0.05$, and the model was extremely well fitted. The CV (coefficient of variation) was 0.8%, and the value of the precision was 27.12 (an adequate precision), which indicated that the model reliability was good. The difference between predicted R^2 and corrected R^2 was less than 0.2, so the model was reliable. Among the parameters, the discharge opening length and the angle of the hopper bias had a very significant effect on the filling coefficient of the granular feed filled into the spiral, and the interaction of the two factors also had a very significant effect on the filling coefficient. In addition, the square of the hopper bias angle also had a significant effect on the filling coefficient. In contrast, the square of the discharge opening length had a non-significant effect on the filling coefficient. Therefore, the model could be improved by deleting the term of the square of the discharge opening length.

Table 5. Results of response surface tests.

Level	Discharge Opening Length (mm)	Inclination Angle (degree)	Filling Coefficient
1	79.2893	8.7868	0.585361
2	220.711	8.7868	0.609171
3	79.2893	51.2132	0.535104
4	220.711	51.2132	0.598109
5	50	30	0.548144
6	250	30	0.624044
7	150	0	0.595142
8	150	60	0.561056
9	150	30	0.584116
10	150	30	0.59386
11	150	30	0.587706
12	150	30	0.594885
13	150	30	0.590142

The analysis of variance for the modified model is shown in Table 7. The fitting results showed that the *p*-value of the model was <0.0001 , and the *p*-value of the lack-of-fit term was $0.3968 > 0.05$. The CV (coefficient of variation) was 0.82%, and the value of the precision was $29.6735 > 4$ (an adequate precision), which indicated a higher reliability of the model. The predictive coefficient of determination R^2 and the corrective coefficient of determination R^2 were 0.9610 and 0.9212, respectively, and the difference between them was less than 0.2,

indicating the model was usable. The model was better than the pre-optimization model in terms of accuracy and reliability.

Table 6. Analysis of variance.

Index	Source of Variance	Square Sum	Degrees of Freedom	Mean Square	F-Value	p-Value
Filling coefficient	Model	0.0069	5	0.0014	62.89	<0.0001
	Discharge opening length	0.0047	1	0.0047	215.9	<0.0001
	Inclination angle	0.0015	1	0.0015	68.71	<0.0001
	$L\theta$	0.0004	1	0.0004	17.6	0.0041
	L^2	0	1	0	1.36	0.2821
	θ^2	0.0003	1	0.0003	11.71	0.0111
	Residual	0.0002	7	0		
	Lack of fit	0.0001	3	0	1.26	0.4004
	Pure error	0.0001	4	0		
	Total	0.007	12			

Table 7. Analysis of variance for the modified model.

Index	Source of Variance	Square Sum	Degrees of Freedom	Mean Square	F-Value	p-Value
Filling coefficient	Model	0.0068	4	0.0017	74.92	<0.0001
	Discharge opening length	0.0047	1	0.0047	206.66	<0.0001
	Inclination angle	0.0015	1	0.0015	65.76	<0.0001
	$L\theta$	0.0004	1	0.0004	16.84	0.0034
	L^2	0.0002	1	0.0002	10.41	0.0121
	θ^2	0.0002	8	0		
	Residual	0.0001	4	0	1.32	0.3968
	Lack of fit	0.0001	4	0		
	Pure error	0.007	12			
	Total	0.0068	4	0.0017	74.92	<0.0001

To construct the mathematical relationship between the discharge opening length, the hopper inclination angle, and the filling coefficient, the regression analysis of the above test results was carried out, and the regression equation between the filling coefficient and the two factors was obtained as shown in Equation (8), which can be used to predict the filling coefficient in the screw conveyor under different coupling parameters. Ultimately, substituting Equation (8) into Equation (4), the screw feeder flow rate model was obtained as Equation (9), which takes into account the effect of the coupling parameters of the hopper to the screw conveyor on the flow rate:

$$\varphi = 0.57440 + 0.00015L - 0.00085\theta + 0.00001L\theta - 0.0001\theta^2 \quad (8)$$

where φ is the material filling coefficient, L is the discharge opening length, in mm, and θ is the hopper inclination angle.

$$Q = 47D^2Sn\lambda\varepsilon(0.57440 + 0.00015L + 0.00085\theta + 0.00001L\theta + 0.0001\theta^2) \quad (9)$$

3.5. Flow Rate Model Validation

According to the influencing factors involved in the model in (8), five spiral feeders were randomly modeled geometrically for the simulation tests. A discharging test bench was set up under laboratory conditions for the actual tests as shown in Figure 8. The combinations of discharge opening length and hopper inclination angle were set to 250 mm and 0°, 150 mm and 15°, 80 mm and 45°, 150 mm and 24°, and 220 mm and 45° for the five screw feeders, respectively. Before the simulation and actual discharging started, the pellet feed was pre-filled to the same height and then discharged at a screw speed of 40 rpm, and the discharging flow rate data were recorded in real time by using an electronic balance (CHS-D; An Heng).

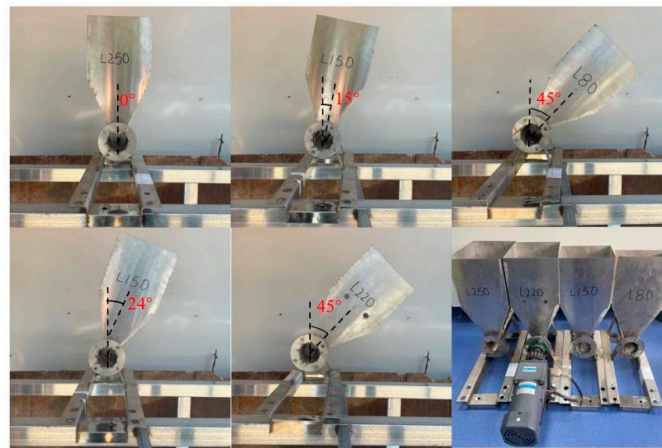


Figure 8. Screw feeder discharging test bench.

Comparing the simulation and actual discharging rate with the predicted discharging rate obtained by solving model (9), which are shown in Figure 9, the results showed that the discharging rate obtained by the prediction model was close to the value obtained from the simulation and the actual test, and the maximum prediction errors were 6.8% and 5.0% compared with those of the simulation and the actual test, respectively. Moreover, when the varying hopper–conveyor coupling parameters (i.e., discharge opening length and hopper inclination angle) led to changes in the actual discharge flow rate, the results calculated by the prediction model could accurately reflect the flow rate varying trends, which cannot be achieved with traditional models (Equation (4)). Therefore, the results of the validation test indicated that the corrected prediction model could predict the screw feeder discharge flow rate, within acceptable margins of error, while considering the hopper–conveyor coupling parameters.

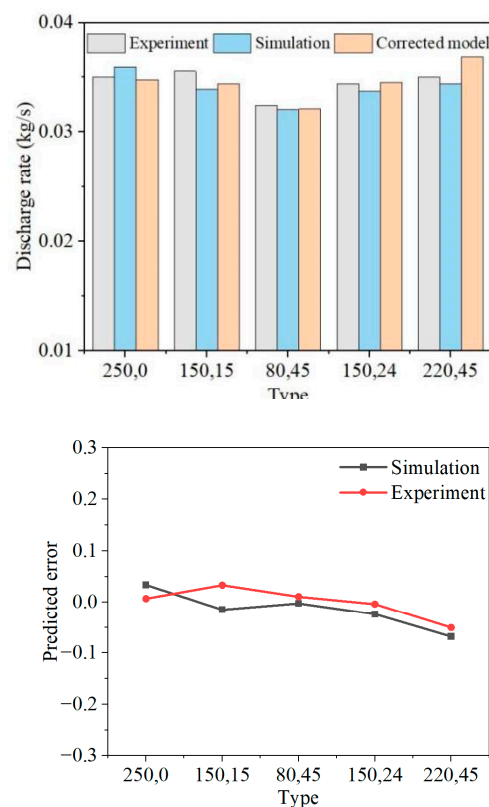


Figure 9. Validation test results of coupled hopper–screw conveyor flow rate modeling.

4. Discussion

A screw feeder always consists of a wedge hopper and a screw conveyor. The flow rate of the hopper and the conveyor can both be predicted by readily available flow rate models independently. But neither model takes the effect of the hopper–conveyor coupling parameters into account. From the results of this study, some of the hopper–conveyor coupling parameters (e.g., hopper discharge opening length and hopper inclination angle) have a significant effect on the flow rate of the screw feeder. Therefore, only by taking these parameters into account when modeling the flow rate can the model predict the change in flow rate due to the variation in the coupling parameters. In this study, the coupling parameters that had significant effects on the flow rate were screened out by using a single-factor test, and then a flow rate model considering the significant coupling parameters was constructed by means of a response surface test. The results of the validation tests showed that the model not only predicted the magnitude of the screw feeder flow rate but also responded to the change in the flow rate due to the change in the coupling parameters. Compared to the traditional empirical formula, the corrected model had a wider range of applicability.

It should be noted that this study still has some limitations. Firstly, the screw feeder involved in this study is used in the field of pig feeding and is mainly aimed at pellet feed delivery. The dimensions of the screw conveyor were determined within the range commonly used in this application scenario. When the size of the screw conveyor is outside of this setup range, the reliability of the model needs to be verified by further tests. Secondly, the material delivered to the screw conveyor was pellet feed, which was represented by a uniform spherical cylindrical in this study. In practice, it is expected that the characteristics of the material will have a great impact on the flow rate model [33,34]. The particle size distribution, particle shape, viscosity, etc., will affect the flow state in the hopper and the filling process in the conveyor. In future studies, one direction to consider is to explore how the coupling parameters affect the material filling coefficient when conveying materials with different particle sizes and shapes, which is important to improve the suitability of the model.

5. Conclusions

In this study, based on the analysis of the flow rate model of a wedge-shaped hopper and screw conveyor, a simulation discharge test of the screw feeder was carried out to study the influence of the hopper discharge opening length, discharge opening width, hopper offset distance, hopper inclination angle, and hopper semi-apex angle on the filling coefficient in the flow rate equation of the screw feeder. A response surface test was carried out to construct the regression model of the filling coefficient, and then the discharge flow rate model was corrected by considering the hopper–conveyor coupling parameters. The main conclusions are as follows:

- (1) The discharge rate of the screw feeder consisting of a wedge-shaped hopper and a screw conveyor is mainly determined by the screw conveyor. Among the coupling parameters of the screw feeder, the hopper discharge opening width, hopper offset distance, and hopper semi-apex angle have no significant effect on the filling coefficient of the pellet feed in the screw, while the hopper inclination angle and the hopper discharge opening length have a significant effect on the filling coefficient.
- (2) A flow rate model considering hopper–conveyor coupling parameters was constructed. The maximum prediction errors were 6.8% and 5.0% compared with the simulation and the actual discharging. Meanwhile, the model could accurately reflect the flow rate varying trends due to the change in the coupling parameters.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/agriculture14071203/s1>, Figure S1: Results of time step independence tests; Figure S2: Results of mesh independence tests; Table S1: Results of time step independence tests and ANOVA analysis; Table S2: Results of mesh independence tests and ANOVA analysis.

Author Contributions: Conceptualization, P.C., T.H., and B.W.; methodology, P.C. and B.W.; software, P.C., T.H., and H.Q.; validation, H.Q., D.L., and X.L.; formal analysis, P.C. and H.Q.; investigation, P.C., B.W., and T.H.; resources, B.W., F.X., and B.L.; data curation, D.L. and X.L.; writing—original draft preparation, P.C. and T.H.; writing—review and editing, B.W.; visualization, P.C. and T.H.; supervision, B.W.; project administration, B.W., F.X., and B.L.; funding acquisition, B.W. and B.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by Changsha Municipal Natural Science Foundation (kq2208071), the Scientific Research Fund of the Hunan Provincial Education Department (23A0181), the 2023 Central Agricultural Machinery Research and Development, Manufacturing, Promotion, and Integrated Application Pilot Fund, China Project (Hunan Finance Department's Pre-(2023) No. 204 Document), the Hunan Provincial Natural Science Foundation of China (2024JJ5194), and the Postgraduate Scientific Research Innovation Project of Hunan Province (CX20230723).

Institutional Review Board Statement: Not applicable.

Data Availability Statement: All data are presented in this article in the form of figures and tables.

Conflicts of Interest: The authors declare no conflicts of interest.

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